Phys1103 Experiments

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Experiment-2

Straight Line Motion with Constant Velocity

In this section of the experiment, you will study and determine the velocity of an object moving in a straight line with constant velocity. In this type of motion, the moving object travels equal distances in equal time intervals along a straight line. The puck moving freely on the surface of the horizontal (leveled off) air table will be considered as the particle. The puck will move with constant velocity across the air table after it is set in the motion with a small push.

Theory

When a particle moves along a straight line, we can describe its position with respect to an origin (**0**), by means of a coordinate such as x. If there is no net force acting on a moving object, it moves on a straight line with a constant velocity. The displacement of the particle is the change in its position. So, the particle's **average velocity** (v_{av}) during a time interval ($\Delta t=t_2-t_1$) is equal to its displacement ($\Delta x=x_2-x_1$) divided by Δt:

$$v_{av} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{\Delta x}{\Delta t}$$ **(Experimental)** (1)

From the Equation-(1), the average velocity is the displacement (Δx) divided by the time interval (Δt) during which the displacement occurs. The direction of the average velocity is always the same as the direction of the displacement. Average velocity does **not** depend on the actual distance that a particle moves (that is, the total distance traveled), but instead **depends on** its original and final position (**displacement**).

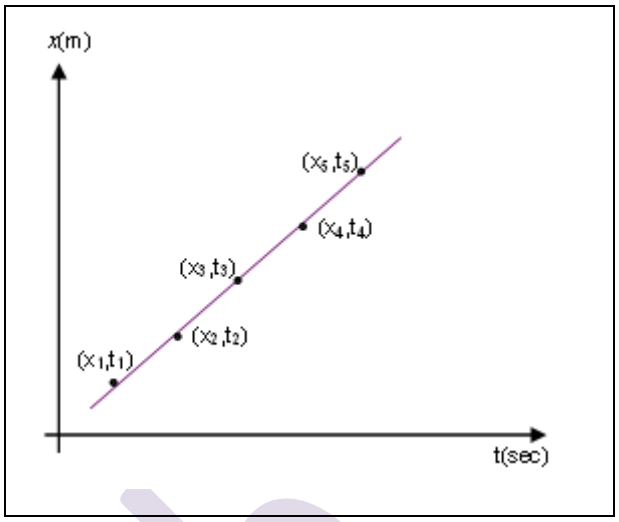


Figure-4: Position as a function of time for an object moving in straight line with constant velocity.

So, the **average velocity** is defined in terms of **displacement**, rather than total distance traveled. The velocity of a particle can also be found from the graph of its **position** as a function of the **time**. On a graph of the **position** (x) versus **time** (**t**), the **average velocity** for a time interval (Δt) is the **slope** of the straight ***line*** connecting the points on the curve that represent the two ends of the time interval. So, in this experiment, if we plot a graph x versus **t** (x-**t** graph), we will have a straight line with a slope and the slope of that line will give the **experimental** value of the average velocity for the motion.

The displacement $x(t)$ of a particle moving in a straight line with constant velocity is given as a function of time in the form:

$$x(t) = x_0 + vt \quad (2)$$

If at **t=0**, the particle was at the **origin**, then $x_0=0$. So, when the object is at the origin with the initial position $x_0=0$, the equation of the motion becomes at any time:

$$x(t) = vt \quad (3)$$

From the above relation, it is obvious that the particle travels equal distance in the equal time intervals along a straight line (see Figure-4).

2.1. Experimental Procedures

Caution!

Do **not** touch the conducting portions, high voltage pins or the hoses of the pucks and the recording or the carbon paper **while spark timer is working**.

1. Construct the experimental set-up as shown in the Figure-(3) without switching on the main power of the **spark-timer**.

   It is noticed that the pucks should be connected to the pump through the hoses. The air hoses should be free and not twisted.

2. Place first the conductive **carbon paper** and then white **recording paper** as the data sheet on the glass plate of the air table.

   2.1. The carbon paper and recording paper should be flat on the air table.
   2.2. Put two pucks on the **recording paper**. **Keep** one of the pucks stationary on a folded piece of data sheet at one corner of the air table. *In this part of the experiment, only **one puck** will be used.*

3. Level the air table **horizontally** by turning on the **compressor** (air supply) and placing one of the pucks at the table's center. Adjust the legs of the air table so that the puck will **not** move.

4. Now, turn on the spark-timer and set the spark timer frequency to:

   - f=20Hz.

5. Test both two switches for the compressor and spark-timer operations. The single puck should move easily, almost without friction when the compressor works only by pressing the **pump** (puck) **foot-switch**. When the **spark-timer foot switch** is pressed, black dots on white paper should be observed (on the side of the recording paper facing the carbon paper).

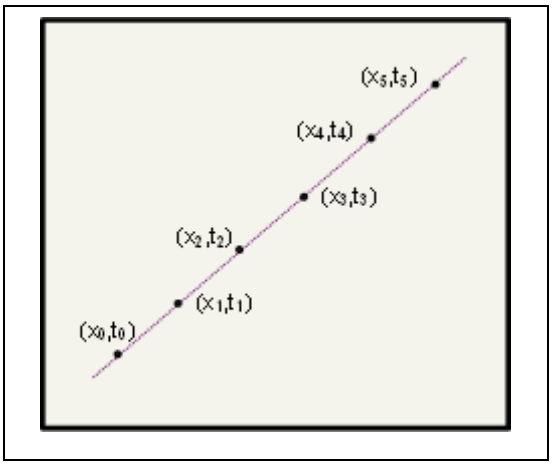


Figure-5: The dots produced by the puck on the data sheet.

6. Then, test the puck motion by activating both the puck **switch** and spark-timer **switch** in the same time and observe the black dots on the recording paper. Make sure that the puck is moving freely on the air table.

7. Activate both compressor (P) and spark timer (S) switches and then, **push** the puck **diagonally** across the surface of the air table.

   7.1. At the moment you **release** the puck, it will move along the whole diagonal distance across the air table in a straight line with **constant velocity**.
   7.2. Then, stop the both switches at the instant the puck reaches its maximum positive displacement, where it will turn back.

8. **Turn off the spark-timer and then remove the puck from the air table for safety.**

9. Remove the white recording paper (data sheet) from the air table.

   9.1. The black dots on the recording paper will look like those given in the Figure-(**5**).
   9.2. The dots made by the spark timer on the recording paper mark the displacement of the puck with equally spaced intervals of **time**.

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10. Number and **encircle** the dots **starting from the first dot** as **0,1,2,….,10**. Take x_0=0 and t_0=0 at the first dot (dot-**0**).

10.1. Using a ruler, **measure** the distances of the **10 dots** starting from **first dot** (**dot-0**).

10.2. The displacement x relative to the initial position (**dot-0**) can be measured directly by using a ruler.

10.3. Find also the **time (t)** corresponding to each dot (that is, consider the **first dot** at time zero and each successive dot as **0.05 seconds** after the previous dot since f=20Hz).

10.4. Record the data values of the position (x) and time (**t**) in the Table-(**1**).

> The time between two dots is 1/20 seconds since the spark timer frequency was set to f=20Hz. So, take the first dot at time zero and each successive dot as "0.05" seconds after the previous dot.

11. Using the data points in Table-(**1**), **plot** the **position** (y-axis) vs **time** (x-axis). The x-**t** graph must show a linear function.

11.1. Label the axes and write down the corresponding **unit** on each axis.

11.2. Draw the best line that fits a linear graph and find the **slope** of the best-line.

11.3. Then, determine the **velocity** (v) of the puck by using the **slope** of the line. The slope of the line is the **average velocity** of the puck.

> The **slope** of the best-fit line on a position-time graph is the average velocity of the object. Average velocity is the rate of change in position for a time interval.

12. Using the data values in Table-(**1**), construct the Table-(**2**).

12.1. Measure the **position** (that is, the **distance** between **2** successive dots) and the **time** corresponding to each **dot interval**.

12.2. Calculate the **average velocity** (v_{av}) from the table for each dot interval.

12.3. Record this on the column marked v_{av} (average speed) in units of **m/sec**.

> Note that the average velocity depends only on the displacement that occurs during the time interval $\Delta t=t_2-t_1$. So, divide each **distance** by the period T=1/f of the spark-timer to obtain the average velocity (v_{av}) of the puck between each pair of dots.

13. **Compare** the **average velocity** for each dot interval with the value which is obtained from the **slope** of graph.

13.1. Fill your data in the Table-(**3**).

13.2. Is the velocity determined by the **slope** consistent with the average velocity calculated for **each dot interval**?. If your comparison shows a percent difference more than 10%, explain at least one of the possible error sources.

Conclusions and Questions

- Discuss the difference in the velocity values calculated for **each dot interval** and the values found from the graph. Is the difference approximately the same?.
- If an object moves at a uniform (that is, constant) velocity during a particular time interval, is its **instantaneous velocity** at any instant the same as its **average velocity**?. Explain your answer.
- Describe how you use a position-time graph to determine average velocity.
- What does a **zero slope** of a position-time graph indicate about the object's average velocity?.
- What is the uncertainty in your measurement of displacements between dots?. Explain briefly the possible sources of the error in the experiment?.

2.2. LABORATORY REPORT

Straight Line Motion with Constant Velocity

Table-1: Data values for the position and time of the motion with **constant velocity**.

Dot Number	Position $x \pm \Delta x(m)$	Time $t(\text{sec.})$	By Slope $v(m/s)$
0	0	0	.....
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

Table-2: Experimental data values for the average velocity of the motion with constant velocity.

Interval Number (n)	$x_n(m)$	$x_{n+1}(m)$	$x_{n+1}-x_n(m)$	$t_n(\text{sec.})$	$t_{n+1}(s)$	$t_{n+1}-t_n(s)$	$v_{av}(m/s)$
0-1	**0**	**0**	**0**	**0**	**0**		
1-2							
2-3							
3-4							
4-5							
5-6							
6-7							
7-8							
8-9							
9-10							

(Note that x_n is the position of the nth data point corresponding to the related dot).



Graph-1: The **position** as a function of **time** for the puck moving in a straight line with **constant velocity**.

Table-3: **Comparison** of the velocity found by the slope with the **average velocity** of the each time interval.

Interval Number (n)	$v_{av}(m/s)$	**By Slope** $v(m/s)$	*Difference*, $\lvert\Delta v(m/s)\rvert$	*Difference* (±%)
0-1	. . . . .	. . . . .	. . . . .	. . . . .
1-2	. . . . .		. . . . .	. . . . .
2-3	. . . . .		. . . . .	. . . . .
3-4	. . . . .		. . . . .	. . . . .
4-5	. . . . .		. . . . .	. . . . .
5-6	. . . . .		. . . . .	. . . . .
6-7	. . . . .		. . . . .	. . . . .
7-8	. . . . .		. . . . .	. . . . .
8-9	. . . . .		. . . . .	. . . . .
9-10	. . . . .		. . . . .	. . . . .